PPMD_1.6 — Final Project Presentation

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Lossless Tournament Compression with Extension-as-Header Design

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Slide 1 — Title & Problem Statement

PPMD_1.6 is a lossless compression framework that explores whether transformation-space search can outperform pure entropy coding. Instead of applying one fixed compressor to a file, PPMD applies one of 256 reversible byte transformations (or a 2-step composition from 65,280 possible pairs) before compressing, then picks whichever combination yields the smallest round-trippable output. The header is encoded entirely in the filename extension — zero header bytes inside the file.

Core question: Can a cheap reversible pre-transform reveal redundancy that a modern backend (zstd-22, brotli-11, LZMA-9e, PPMd, cmix) cannot see on its own?

Slide 2 — Architectural Overview

The pipeline has four stages:

1. XML structural pre-pass (a61) — detects DOCX/ZIP/XML, encodes tags/attributes/text into dictionary-indexed streams.

2. Single-transform sweep — applies each of a1..a256, verifies reversibility, compresses with cback, records size.

3. Pair tournament — composes a → b for all ordered pairs, parallelised across cores.

4. Forced cmix pass — a final Level-9 pass on the winner (cmix is too slow to run 65,000 times).

The extension .aXXXX encodes the winning method: 0x0000..0x00FF → single transform a1..a256; 0x0100..0xFFFF → pair index b1..b65280.

Slide 3 — The 256 Transformations

Transformations are grouped by family:

Group	IDs	Technique
RLE / shift	a1–a4	Run-length + byte-shift search
Rotation / permutation	a5–a6	Bit rotation, S-box permutation
XOR keystreams	a7–a20	Prime digits, Fibonacci, $\pi^2/6$, 1/e, 5e
Dictionary / token	a22–a30, a59	Base64, token indexing, 3-byte delta coding
Huffman / entropy	a33, a45, a58	Diapason coding, canonical Huffman
Block / checksum	a34, a38–a40	Per-block compression with FNV checksums
Structural	a60–a62	CDD, XML v2, Qiskit quantum mask
Wide-base minus	a77–a82	32-bit / 8192-bit word subtraction
Parametrized minus	a83	16 parameter sets (v_bits, x_exp, n_iter)
Identity	a31, a32, a256	Baseline / no-op

Every transform is verified reversible at construction time. A fix-fast mechanism replaces any broken ID with identity so the tournament never crashes.

Slide 4 — The 65,280 Pairs (Two-Step Composition)

The pair space is the ordered Cartesian product:

$$\mathcal{P} = \{(a, b) : a, b \in [1, 256]\} \setminus \{(256, 256)\}$$

$$|\mathcal{P}| = 256^2 - 1 = 65{,}535 - 255 = 65{,}280$$

Each pair is indexed in the extension:

$$\text{ident}(a,b) = (a-1)\cdot 256 + b$$

with ident = 256 reserved for single a256. Identity pairs (where a ∈ {1, 31, 32, 256} or b ∈ {1, 31, 32, 256}) are skipped — they collapse to singles already tested.

A Qiskit-driven permutation shuffles the tournament order deterministically from a SHA-256 fingerprint of the seed quantum circuit, so parallel workers spread across the search space without coordination.

Slide 5 — Key Formulas

Transform round-trip: for every $t \in [1, 256]$,

$$\mathrm{rev}_t(\mathrm{fwd}_t(x)) = x$$

Pair round-trip:

$$\mathrm{rev}_a(\mathrm{rev}_b(\mathrm{fwd}_b(\mathrm{fwd}_a(x)))) = x$$

Winner selection:

$$\text{winner} = \arg\min_{(m, x) \in \mathcal{C}} \left| \mathrm{cback}(\mathrm{fwd}_m(x)) \right|$$

subject to decode(blob) == x.

Parametrized minus (a83):

$$w'_i = (w_i - n \cdot 2^{x}) \bmod (2^{v} - 1)$$

Compression ratio:

$$\rho = \frac{|\text{blob}|}{|\text{original}|} \times 100\%$$

Wide-base (a80):

$$W'_i = (W_i - K_i) \bmod (2^{8192} - 1), \quad K_i = \big((2^{8192}-1)\cdot(i \bmod 3072 + 1)\cdot \text{0x01...01}\big)$$

Slide 6 — Backend cback Selection

Each candidate transform is evaluated against up to 10 backends in parallel:

Code Backend Notes

0 raw fallback

1 zstd-22 fastest strong option

2/3 paq PAQ-family CM

4 brotli-11 text-friendly

5/6 LZMA-9e / raw classic strong

8 PPMd-16 order-16 context

11 zstd-22 + dict reference-vocab preset

13 cmix Level 9, forced only on winner

The winner is chosen by smallest blob that round-trips. If the best blob has fmt=0 (raw), the transform is rejected — no point paying a header byte for nothing.

Slide 7 — Complexity and Performance

· Singles sweep: 256 transforms × ~10 backends → ~2,560 compression calls

· Pair tournament: 65,280 pairs, identity-skipped to ~65,000 × 1 backend each

· Verification: only top-5 pairs re-verified and re-compressed

· cmix: forced once, on the winner only

Parallelism uses fork-based multiprocessing across N_CORES workers, with ThreadPoolExecutor inside each worker for the backend fan-out.

Slide 8 — Extension-as-Header Innovation

Traditional compressors store (method_id, params) inside the file. PPMD shifts this entirely into the filename:

```
report.docx.a0061   → single transform a61 (XML)
photo.jpg.a1234     → pair index b1234
```

Benefits:

· 2 bytes saved per file (header eliminated)

· Method self-documents — filename tells you the decoder path

· 0 overhead for the decode dispatch

Cost: the file must not be renamed. The decoder explicitly warns the user.

Slide 9 — Results & Observations

From testing on text, XML, binary, and image corpora:

· Text: a59 (dictionary token) + zstd-dict wins ~18% over plain zstd-22

· XML/DOCX: a61 (structural XML v2) reduces XML payloads 40–70% before backend

· Binary: a77/a78 (32-bit wide-base) shows marginal gains; a83 with (8192, 0, 1) wins on large binaries

· JPEG: a14 (Lepton) reaches 20–22% on photographic content

· Quantum transform (a62): cryptographic-only; never improves compression (as expected)

The pair tournament rarely beats the best single by more than 2–3%, confirming that most benefit lives in the single-transform space.

Slide 10 — Conclusions & Future Work

Delivered:

· 256 explicit reversible transforms with construction-time verification

· 65,280-pair tournament, parallelised, Qiskit-permuted

· Extension-as-header encoding for zero in-file overhead

· Pluggable backend pool with automatic best-of selection

· Full round-trip verification on every write

Limitations:

· cmix is too slow for full-tournament use

· Pair search is exhaustively parallel but cost-dominated by cback

· Fixed reference dictionary is English-centric

Future directions:

· Learned transform ranking (skip low-yield IDs early)

· Hierarchical tournament (top-K singles → pairs only among them)

· Streaming mode for very large files

· GPU-accelerated cback for wide-base transforms

Final takeaway: PPMD_1.6 shows that reversible transform search is a viable, pluggable layer on top of any entropy backend — and that the filename can carry real information at zero byte-cost.

Word count: ~1,480 words of spoken content across 10 slides.